
SAME EVIDENCE, DIFFERENT SKILL: DIAGNOSING STATE-GENERATION VARIANCE IN SELF-EVOLVING AGENTS

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ABSTRACT

Self-evolving agents convert execution feedback into persistent skills, but most studies treat the state-writing step as an implementation detail and focus instead on which trajectories should be shown to the updater. We ask whether this misses a distinct failure mode: the same evidence can be converted into materially different persistent states. We begin with a controlled DeepSeek-V4-Pro study of 48 matched evidence pairs, 96 learned states, and 1,728 held-out evaluations. Replacing winner evidence with a mixed rejected-witness view produces only a small heterogeneous mean gain of 0.023, with a frozen confidence interval that crosses zero. A single-stream mechanism study then separates evidence from state realization. A historically strong First-Fail state remains useful when frozen and re-evaluated, whereas two fresh updater realizations from reconstructed byte-identical evidence fail to reproduce that advantage. This combination localizes a candidate bottleneck at persistent-state generation rather than evidence availability alone, while leaving broad failure-evidence superiority unsupported. We formalize the pipeline as evidence selection $\rightarrow$ state generation $\rightarrow$ persistent state $\rightarrow$ downstream behavior and introduce a prospective variance-controlled typed compiler that maps learner-visible trajectory+score packages to a finite repair representation and canonical skill state. The planned bridge crosses WINNER versus First-Fail-4 evidence with native free-form versus constrained compilation on fresh development streams, including a score-matched trajectory-blind generic control. The present evidence supports a state-generation bottleneck hypothesis and a concrete causal test for it; it does not yet establish that the compiler improves utility or that rejected failures are uniquely beneficial.

1 INTRODUCTION

Self-evolving agents improve without updating model weights by turning execution experience into persistent procedural state. A spreadsheet agent, for example, may execute several attempts of the same task, retain successful or failed trajectories, and ask an LLM updater to revise a reusable skill document for future tasks. This paradigm is attractive because it moves adaptation into an inspectable persistent artifact rather than another round of model training. Recent systems already show that failed trajectories can matter, that success/failure contrasts can reveal useful corrections, and that accumulated execution knowledge can support later skill evolution (Liu et al., 2026b; Chen et al., 2026; Liu et al., 2026a; Tang et al., 2026).

A common assumption nevertheless remains implicit: once the updater is given the “right” evidence, writing the persistent state is mostly a downstream implementation step. Under this view, the scientific question is which trajectory or feedback view should reach the learner. Our experiments challenge that reduction. In a controlled study with the same actor, verifier, updater implementation, initial skill, and held-out evaluator, replacing winner evidence with a rejected-witness view does not yield a stable population-level advantage. More importantly, in a selected development stream, a persistent state generated from First-Fail evidence remains useful when frozen, yet fresh updater runs from reconstructed byte-identical evidence fail to recreate the same advantage. The evidence did not disappear; the useful state realization did.

This distinction matters because persistent skill text is itself an intervention on future behavior. Two updater calls can receive the same evidence and still synthesize states that differ in scope, redundancy, ordering, specificity, or procedural commitments. If downstream utility is sensitive to those differences, then “which evidence was shown?” and “how was that evidence compiled into state?” are separate causal variables. A self-evolution pipeline can therefore fail even when evidence selection is held fixed.

We study this possibility through the decomposition

$$\text{search trajectories} \rightarrow \underbrace{E}_{\text{evidence selection}} \rightarrow \underbrace{G}_{\text{state generation}} \rightarrow \underbrace{S}_{\text{persistent state}} \rightarrow \underbrace{Y}_{\text{future utility}}. \quad (1)$$

The missing scientific object is the *state-generation realization*: the persistent artifact produced from a fixed learner-visible evidence package. This object is not equivalent to trajectory choice, state semantics, or downstream actor randomness. Identifying it requires holding some of those layers fixed while varying or freezing others.

Our evidence follows that logic rather than a chronology of increasingly complex methods. First, a 48-pair controlled study shows that rejected-witness evidence is not uniformly superior to winner evidence. Second, a single-stream witness-selection intervention fails to recover a simple “more diagnostic failure is better” law. Third, the historically strong First-Fail state remains directionally useful under two frozen-state remeasurements, weakening a pure downstream-actor-noise explanation. Fourth, two fresh updater realizations from byte-identical First-Fail evidence fail to reproduce the historical advantage. Together these observations do not prove a general variance decomposition, but they localize a concrete candidate bottleneck at the state-generation step.

This localization motivates a narrower method than generic skill revision. We introduce *variance-controlled typed state compilation*: instead of asking a free-form LLM writer to synthesize the entire persistent skill, the learner maps visible trajectory+score evidence into a small typed diagnosis and canonical repair vocabulary, then compiles the selected primitives deterministically into the skill. The point is not that structured diagnosis itself is novel—SkillRevise already performs execution-grounded diagnosis and repair, SkillCAT uses same-task success/failure contrasts, and WikiSkill separates raw experience from persistent knowledge (Liu et al., 2026a; Chen et al., 2026; Tang et al., 2026). Our residual question is whether free-form *state synthesis itself* is a bottleneck under matched evidence, and whether controlling that interface changes reproducibility and downstream utility.

We therefore preregister a fresh development bridge that independently crosses evidence source and state-generation method: WINNER versus First-Fail-4 evidence, and native free-form updating versus constrained typed compilation. The design uses disjoint SCREEN and VALIDATION panels and includes a score-matched trajectory-blind generic compiler control, so a positive result cannot be attributed merely to seeing failure labels or injecting generic spreadsheet hygiene. This bridge has not yet produced scientific outcomes; we keep it prospective rather than using manuscript language to promote a method result that does not exist.

Contributions. We make three current contributions and state one prospective test explicitly.

- **A causal decomposition of persistent self-evolution.** We separate evidence selection, state generation, persistent state, and downstream behavior, making state realization an explicit object of analysis.
- **A localization result.** Across controlled large-sample and single-stream interventions, we find that rejected-witness evidence is heterogeneous, a strong frozen state is behaviorally stable, and byte-identical evidence does not reliably regenerate the same useful state. This supports a candidate state-generation bottleneck without establishing broad failure-evidence superiority.
- **A variance-controlled compiler interface.** We formalize a finite typed diagnosis/repair representation with canonical compilation, content-addressed state identity, and no access to hidden arm, family, or task labels.
- **A prospective falsification experiment.** A fresh 2×2 Evidence $\times$ Generator bridge and score-only generic control are frozen to determine whether the compiler effect, the First-Fail evidence effect, and their interaction survive outside the selected mechanism case.

2 PROBLEM SETUP AND CLOSEST WORK

2.1 PERSISTENT SKILL SELF-EVOLUTION

For an update stream u , the agent executes eight tasks. Task i produces a frozen search pool $P_i = \{\tau_{ij}\}_{j=1}^K$ with binary verifier scores $r_{ij} \in \{0, 1\}$. A serving rule chooses an acting winner for the user-facing task, while an evidence projection E determines which trajectory+score package is exposed to the persistent learner. The state generator G maps the eight learner-visible evidence units and initial skill S_0 to a new persistent skill,

$$S = G(E(P_1), \dots, E(P_8); S_0). \quad (2)$$

A frozen downstream actor then executes held-out task x conditioned on S , producing utility $Y(x; S) \in \{0, 1\}$ under the deterministic spreadsheet verifier.

This decomposition creates three distinct estimands. The *evidence-package effect* compares different E while holding the generator fixed. The *state-generation-method effect* compares different G on the same selected evidence. The *state effect* holds the state bytes fixed and remeasures downstream behavior. These are not interchangeable. In particular, a useful frozen state does not imply that the updater can reliably regenerate it, and an evidence source that sometimes yields a useful state does not imply that the evidence source is generally superior.

2.2 CONTROLLED SPREADSHEET SUBSTRATE

Our mechanism studies use a locally generated SpreadsheetBench-compatible suite with six predeclared failure families: input/output contract, target sheet/range, schema/key alignment, aggregation/join, formula/materialization, and multi-step pipeline. Each task has deterministic initialized and golden workbooks and a deterministic verifier. This suite is used for causal control, not as a public benchmark. SpreadsheetBench itself contains real-world spreadsheet manipulation tasks and OJ-style evaluation (Ma et al., 2024); external benchmark claims are deliberately deferred until the internal mechanism survives.

The primary actor and updater model in the current mechanism line are DeepSeek-V4-Pro under a frozen Ark route. Actor decoding uses temperature zero, thinking disabled, and no provider retry. The first-party persistent updater is the MindMemOS SkillEvolver path used throughout the controlled study. Exact model routes, skill hashes, trajectory hashes, budgets, and task IDs are bound in execution receipts rather than inferred from manuscript prose.

2.3 CLOSEST WORK AND RESIDUAL SCIENTIFIC OBJECT

Failure-aware feedback and skill evolution. Rethinking Self-Evolving Agent Skills varies the feedback presented to a skill optimizer and shows that failure-containing feedback is often selected, but the ranking of Normal, Fail-only, and Success-only conditions depends on model and benchmark (Liu et al., 2026b). This directly rules out a novelty claim that failures can help. Our residual question is different: with evidence bytes fixed, can the state-writing step itself become an unstable bottleneck?

Contrastive trajectory diagnosis. SkillCAT samples multiple trajectories, contrasts same-task successes and failures, extracts candidate corrections, and validates patches before merging (Chen et al., 2026). It therefore already covers success/failure pairing and contrastive causal extraction. Our First-Fail evidence arm is not presented as a new contrastive-learning idea. We instead use matched evidence arms to factor evidence selection from the generator that materializes persistent state.

Execution-grounded skill revision. SkillRevise diagnoses skill defects from execution traces, retrieves reusable repair principles, and applies execution-anchored revisions that are re-executed and selected by utility (Liu et al., 2026a). Consequently, “trajectory $\rightarrow$ structured diagnosis $\rightarrow$ skill edit” is not our contribution. Our typed compiler is scientifically useful only if it tests a different claim: whether replacing unconstrained state synthesis with canonical compilation changes state reproducibility or downstream utility under the same evidence.

Persistent knowledge accumulation. WikiSkill separates raw execution experience, an accumulated persistent wiki, and executable skills, and uses the knowledge store to guide future skill evolution (Tang et al., 2026). We therefore do not claim novelty for compiling experience into persistent knowledge. The missing object here is narrower: the realized persistent state as a causal intermediate whose variability can be studied independently of evidence selection.

Residual novelty boundary. The paper’s novelty target is therefore not a new feedback source, diagnosis vocabulary, memory store, or validation-gated text editor. It is the combination of (i) identifying state-generation realization as a distinct bottleneck candidate under matched evidence, and (ii) prospectively crossing evidence source with state-generation method so that the two explanations are no longer confounded. If the prospective generator contrast fails, the method-positive story must stop. If the compiler helps but First-Fail evidence does not, the failure-learning claim must be dropped while the state-compilation result may remain.

3 LOCALIZING THE CANDIDATE BOTTLENECK

We use a sequence of interventions that progressively hold more of the pipeline fixed. The goal is not to turn one selected positive case into a general claim; it is to determine which layer remains plausible after simpler explanations fail.

3.1 EVIDENCE-SOURCE EFFECTS ARE SMALL AND HETEROGENEOUS AT SCALE

Our closed DeepSeek V2 study contains 48 matched pairs, 96 learned states, and 1,728 held-out task evaluations. The two primary states differ in the evidence projection exposed to the same frozen updater. Across pairs, the mean MRW–WIN-C utility difference is $+0.0231$, the median is $+0.0139$, and the standard deviation is 0.0780 . The frozen one-sided sign-flip test gives $p = 0.1719$, and the bootstrap confidence interval is $[-0.0185, 0.0660]$.

This study therefore does not establish rejected-witness superiority. Individual streams span positive, null, and negative effects, including values from -0.125 to $+0.181$. The appropriate inference is heterogeneity, not a failed attempt that can be repaired by silently adding samples or widening a threshold.

3.2 SELECTING A MORE “DIAGNOSTIC” FAILURE DOES NOT SOLVE THE PROBLEM

Post-hoc development analysis suggested that progress-related features correlated with pairwise effects, motivating a single-stream witness-selector pilot on `el-tsr-00`. This stream was chosen after outcome inspection and is therefore mechanism development only. We froze four evidence arms: WIN-C, First-Fail, Progress-Fail, and Progress-Contrast. On 18 held-out tasks they score 13/18, 17/18, 14/18, and 14/18 respectively.

Although the historical First-Fail state is strong, the preregistered purpose of this pilot was to test whether a progress-based selector provides a stable repair rule. It does not. The frozen verdict is `S1_SIGNAL_FAIL_STOP_NO_S2`. We therefore reject the simple story that “choose a later or more diagnostic failure” explains the earlier heterogeneity.

3.3 THE STRONG PERSISTENT STATE REMAINS USEFUL WHEN FROZEN

The next intervention holds the persistent state bytes fixed and re-runs only downstream evaluation. The historical strong First-Fail state is compared with the corresponding WIN-C state in two fresh complete actor replicates on the same 18-task development panel. Replicate 1 gives 15/18 versus 14/18; replicate 2 gives 16/18 versus 12/18.

These remeasurements do not make the selected stream representative, and they do not prove that First-Fail evidence caused the state. They do weaken a simpler explanation: the original 17/18 First-Fail result was not merely one lucky downstream actor realization attached to an otherwise inert state. The frozen artifact itself carries behaviorally relevant information.

Table 1: Evidence ladder. The independent unit and scientific interpretation change across stages; no row is promoted beyond its design.

Stage	Held fixed	Observation	Supported interpretation
V2 48-pair study	actor/updater configuration, design	con- task mean MRW–WIN-C = +0.0231, CI crosses 0	evidence-source effect is heterogeneous
S1 selector pilot	selected stream, up-dater path	13/18, 17/18, 14/18, 14/18	simple progress selector fails
Frozen-state remeasure	persistent bytes	state FF: 15/18, 16/18; WIN-C: 14/18, 12/18	strong state has reproducible local behavioral value
Exact-evidence replay	learner-visible evidence bytes	evi- FF: 15/18, 11/18; WIN-C: 15/18, 12/18	same evidence does not reliably regenerate strong state

3.4 BYTE-IDENTICAL EVIDENCE FAILS TO REGENERATE THE SAME ADVANTAGE

We then reconstruct the exact S1 First-Fail evidence rendered to the updater. All 16 rendered packet hashes match the historical inputs. From this byte-identical evidence we create two new free-form updater realizations, then evaluate each fresh state against a fresh WIN-C realization on the same 18 tasks. Replicate 1 is 15/18 versus 15/18; replicate 2 is 11/18 versus 12/18. The mean fresh contrast is -0.0278 .

The contrast with frozen-state remeasurement is the key localization result. Holding the strong state fixed preserves a directional advantage; holding the evidence fixed but regenerating the state does not. Because the fresh updater states and their downstream evaluations both vary, this is not a full variance-component decomposition. It is nevertheless incompatible with the claim that evidence availability alone explains the strong historical state.

3.5 WHAT IS AND IS NOT LOCALIZED

The current evidence supports the following bounded statement:

In one controlled development case, identical trajectory evidence did not reliably regenerate the same useful persistent skill state, while a previously generated strong state retained downstream utility when frozen. Persistent-state generation is therefore a candidate bottleneck that cannot be reduced to evidence availability alone.

It does *not* establish that rejected failures are generally better than winners, that free-form updater variance dominates actor variance in the population, or that deterministic compilation is already superior. These stronger claims require prospective interventions rather than retrospective interpretation of the selected case.

4 VARIANCE-CONTROLLED TYPED STATE COMPILATION

The localization result motivates an intervention on G , the state-generation interface. The goal is not to give the compiler privileged task semantics. It receives the same learner-visible trajectory text and selected binary verifier score as the free-form updater arm, but replaces open-ended state writing with a finite diagnosis and canonical compilation path.

4.1 DESIGN REQUIREMENTS

A valid generator intervention must satisfy four requirements.

R1: Information parity. The compiler may read only the selected trajectory text and score exposed to the corresponding free-form arm. It cannot condition on the experimental arm, projection name, stream/family label, task family, golden answer, held-out result, historical S1 outcome, or hidden provenance. SHA identifiers are retained only for integrity receipts.

R2: Explicit intermediate representation. The method must expose what it inferred before changing the skill. Otherwise a different free-form prompt merely moves the same uncontrolled synthesis step. We therefore represent each evidence unit by

$$d = (\text{failure stage, failed invariants, observed evidence, required repairs}). \quad (3)$$

R3: Canonical state construction. Identical diagnosis bundles must produce byte-identical persistent skills. Duplicate evidence must not increase state length, primitive ordering must be fixed, and redundant repair surfaces must be removed deterministically.

R4: Strong generic falsification. A positive compiler result must exceed a trajectory-blind control that receives the same selected score pattern. Otherwise the apparent benefit could be explained by “a failure happened, so inject generic spreadsheet hygiene” rather than trajectory-conditioned diagnosis.

4.2 TYPED DIAGNOSIS FROM LEARNER-VISIBLE EVIDENCE

The current compiler uses three repair primitives:

$$\mathcal{R} = \{\text{VERIFY_OUTPUT}, \text{COMPLETE_WORKFLOW}, \text{RECOVER_TOOL_ERROR}\}. \quad (4)$$

The vocabulary is intentionally small and frozen for the current development program. For a selected failed trajectory, visible signals map to repairs as follows:

- no materialized write or output save $\rightarrow$ COMPLETE_WORKFLOW;
- output saved but no visible reload/verification $\rightarrow$ VERIFY_OUTPUT;
- visible tool error without visible clean recovery $\rightarrow$ RECOVER_TOOL_ERROR.

A selected successful trajectory is not automatically labelled as needing a generic repair. Multiple repairs may compose.

These rules are deliberately interpretable and narrow. They should not be read as a universal ontology of agent failure. Their scientific role is to instantiate a controlled state generator whose output variability is structurally bounded.

4.3 CANONICAL COMPILER

For diagnoses $D = \{d_i\}_{i=1}^8$ and base skill S_0 , compilation computes

$$C(D, S_0) = S_0 \oplus \text{Canon}\left(\bigcup_i d_i.\text{required_repairs}\right). \quad (5)$$

The canonicalization operator uses a fixed primitive order and deduplicates repeated requirements. COMPLETE_WORKFLOW already contains save, reload, and target verification semantics, so a separately rendered VERIFY_OUTPUT block is omitted when both are requested. RECOVER_TOOL_ERROR composes after completion semantics. Every diagnosis bundle and compiled skill is content-addressed.

The current zero-provider implementation passes tests for deterministic compilation, duplicate suppression, hidden-label exclusion, and the three diagnosis routes. For diagnoses corresponding to the manually frozen mechanism interventions, it reproduces the G1 verification, G2 completion, and G3 completion+recovery skill surfaces byte-for-byte. This is an implementation qualification, not evidence that those states improve future utility.

4.4 SCORE-ONLY GENERIC CONTROL

The strongest simple alternative is that trajectory content is unnecessary: a system could observe only that some selected evidence failed and inject the most robust generic checklist available in the same primitive vocabulary. We therefore freeze SCORE_ONLY_GENERIC_MAX. It receives the same ordered eight binary scores but no trajectory text. If all scores are one,

it returns S_0 . If any score is zero, it emits the maximal nonredundant generic v1 bundle, COMPLETE_WORKFLOW+RECOVER_TOOL_ERROR.

Consequently, $FF4_COMP > SCORE_ONLY_GENERIC_MAX$ is required before we attribute a compiler benefit to trajectory-conditioned diagnosis. Failure of this falsifier does not imply that canonical state construction is useless; it specifically blocks the stronger trajectory-conditioned repair claim.

4.5 WHAT THE METHOD COMPARISON IDENTIFIES

The free-form and compiler paths are complete state-generation methods. They differ in more than determinism: the compiler has a finite ontology, canonicalization, bounded redundancy, and no generative writer. Therefore a positive COMPILER – FREE contrast should be interpreted as a *complete state-generation-method effect*, not an isolated causal effect of determinism or constraint. State length, delta tokens, and number of repair clauses are recorded as mechanism diagnostics rather than post-hoc matched away.

5 EVALUATION CONTRACT

The remaining experiment program is staged so that a negative mechanism result cannot be rescued by opening a larger benchmark or changing the method after outcomes.

5.1 M2: DETERMINISTIC STATE-SEMANTIC INTERVENTION

Before testing an automatic compiler, we ask a narrower sufficiency question on the selected development case. Four frozen persistent states share the same base spreadsheet skill:

- G0: the exact initial skill;
- G1: G0 plus output save/reload/target-cell verification;
- G2: G0 plus a complete Inspect→Read→Compute→Write→Save→Reload/Verify loop;
- G3: G2 plus clean recovery from malformed or failed tool calls.

No updater is called. Each state is evaluated on the same 18 development held-out tasks with DeepSeek-V4-Pro, temperature zero, thinking disabled, $K = 1$, ten turns, and the same deterministic verifier. The primary screen selects the simplest $G1 < G2 < G3$ whose utility exceeds G0 by at least $1/18$. Historical S1 outcomes are excluded from the gate.

This experiment is a state-level mechanism intervention only. G1–G3 were manually constructed after observing the selected strong state and therefore cannot establish automatic learning from failures. If the 18-task primary passes, a pre-frozen 17-task sensitivity excluding the one task split across provider quota eras must remain strictly positive; this sensitivity may veto but never rescue the primary result.

5.2 M3: FROZEN-STATE STABILITY

Only if M2 survives both gates do we remeasure G0 and the simplest selected arm in exactly two fresh complete actor replicates. No additional replicate is allowed because an observed result is inconvenient. Failure to retain a directional advantage stops the state-semantic branch before automatic bridge execution.

5.3 M4: PROSPECTIVE EVIDENCE×GENERATOR BRIDGE

The automatic bridge uses a newly generated development suite that is disjoint from the selected S1 panel and from the untouched E3 confirmation lane. It contains 120 formal task artifacts: 96 update tasks arranged as 12 eight-task streams, 12 SCREEN held-out tasks, and 12 different VALIDATION held-out tasks. SCREEN and VALIDATION each contain one stream per failure family and their held-out panels are disjoint.

For each stream we freeze the 2×2 :

Evidence source	Native free-form updater	Typed compiler
Winner	W_FREE	W_COMP
First-Fail-4	FF4_FREE	FF4_COMP

Winner evidence. All eight update pools expose the served winner trajectory+score package.

First-Fail-4 evidence. A pool is eligible only if it contains both success and failure. A stream is bridge-eligible only with at least four mixed pools. Exactly four eligible pools are selected by a pre-frozen content hash; those expose the deterministic first failed nonwinner, while the other four expose the same winner package as the Winner arm. This equalizes the *replacement count*, not information content or token length.

Information parity. Within an evidence source, free-form and compiler generators receive the same selected trajectory bytes and binary score. The compiler cannot read hidden arm, family, task, rollout, or held-out labels. Before execution, corresponding evidence-input hashes must match.

Stochasticity. The free-form updater and actor both use temperature zero, thinking disabled, and no provider retry. Because the provider does not expose a reliable user-controlled seed for every call, the main bridge estimates the performance of the first prospectively drawn deployed free-form state versus deterministic compilation; it does not claim seed-level determinism. No second updater draw may be requested because the first result is unfavorable.

5.4 PRIMARY BRIDGE CONTRASTS AND STOP RULES

For stream-level held-out utility J , define

$$G_F = J(\text{FF4_COMP}) - J(\text{FF4_FREE}), \quad (6)$$

$$S_C = J(\text{FF4_COMP}) - J(\text{W_COMP}), \quad (7)$$

$$I = [J(\text{FF4_COMP}) - J(\text{FF4_FREE})] - [J(\text{W_COMP}) - J(\text{W_FREE})]. \quad (8)$$

SCREEN passes only if mean $G_F > 0$, at least 4/6 streams have $G_F > 0$, mean $S_C > 0$, at least 4/6 streams have $S_C > 0$, mean $I > 0$, FF4_COMP exceeds SCORE_ONLY_GENERIC_MAX, and no information-parity or technical-integrity failure occurs. VALIDATION remains sealed unless SCREEN passes and repeats the frozen primary rules on six different streams and a disjoint held-out panel.

The decision table is intentionally asymmetric. If G_F fails, we stop the state-generation-method story. If G_F passes but S_C fails, we drop the rejected-failure claim and retain only a possible reliable-compilation contribution. If FF4_COMP fails the trajectory-blind generic control, we drop trajectory-conditioned diagnosis and interpret any gain as generic/canonical state construction. Only a frozen SCREEN+VALIDATION pass can motivate a separate untouched E3 proposal.

5.5 SCIENTIFIC UNITS AND NON-CLAIMS

The primary bridge unit is the independently updated stream, not the 12 individual held-out tasks or provider calls. The controlled suite is a mechanism substrate, not a public benchmark. No second backbone, public benchmark, or E3 confirmation is opened to rescue a negative bridge result. These stages are separate claim expansions that require their own frozen contracts.

6 CURRENT RESULTS AND OPEN CLAIM GATES

At this manuscript checkpoint, M0–M1 are complete and outcome-readable; M2 is an active frozen measurement object whose incomplete outcomes remain sealed; M3–M4 are conditional prospective stages. We report only completed evidence here.

Evidence source alone is unresolved. 48-pair MRW – WIN-C is small and heterogeneous.

↓

Simple witness selection fails. Progress-based selectors do not recover the strong First-Fail case.

↓

Freeze the state. The historical strong state remains directionally useful across two fresh actor remeasurements.

↓

Freeze the evidence, regenerate the state. Two byte-identical-evidence updater realizations fail to recover the historical advantage.

↓

Candidate bottleneck. Persistent-state generation remains a live causal explanation and motivates a prospective generator intervention.

Figure 1: Localization logic. Each step removes a simpler explanation but has a narrower estimand than a full population variance decomposition.

6.1 REJECTED-WITNESS EVIDENCE IS NOT A STABLE UNIVERSAL TREATMENT

The 48-pair DeepSeek V2 study does not pass its frozen superiority criterion. The mean rejected-witness-minus-winner contrast is $+0.0231$, with bootstrap interval $[-0.0185, 0.0660]$ and one-sided sign-flip $p = 0.1719$. The stream distribution includes positive, null, and negative effects. This directly narrows the paper: the central contribution cannot be “failed trajectories are better learning data.”

6.2 A DIAGNOSTIC-WITNESS SELECTOR FAILS ITS DEVELOPMENT GATE

On the selected `el-usr-00` stream, First-Fail is 17/18 while WIN-C is 13/18, but progress-based failure and progress-contrast states are both 14/18. Because the prospective question was whether a more progress-diagnostic selector repairs the instability, the result is a failure of that hypothesis rather than evidence for First-Fail as a general rule. No S2 selector expansion is opened.

6.3 STATE FREEZING AND EVIDENCE REPLAY POINT TO DIFFERENT LAYERS

The historical strong First-Fail state remains directionally stronger than its WIN-C comparator when both states are frozen: 15/18 versus 14/18 in one complete remeasurement and 16/18 versus 12/18 in a second. The same pattern does not survive state regeneration from byte-identical First-Fail evidence. Fresh updater realization 1 yields 15/18 versus 15/18; realization 2 yields 11/18 versus 12/18.

Figure 1 summarizes the logic. The current evidence does not quantify a population variance component, but it separates two statements that were previously conflated: a persistent artifact can be behaviorally useful, while the updater that produced it can still fail to reproduce that artifact’s utility from the same evidence.

6.4 THE STRONGEST CURRENT CLAIM

The strongest manuscript-level claim supported today is deliberately local:

The usefulness of persistent skill self-evolution is not determined by trajectory evidence alone. In a controlled development case, a useful frozen state remains behaviorally active, but reconstructed byte-identical evidence does not reliably regenerate the same advantage through the native free-form updater. This localizes persistent-state generation as a candidate bottleneck that warrants an independent prospective intervention.

The manuscript does not currently claim that the typed compiler improves downstream utility, that First-Fail-4 is superior to winner evidence, that the candidate bottleneck generalizes beyond the development substrate, or that the three repair primitives form a generally sufficient skill language.

6.5 WHAT CHANGES AFTER THE NEXT GATES

If M2 and M3 fail, the state-semantic sufficiency branch stops and the compiler experiment is not promoted. If they pass but M4 shows no compiler advantage, the state-generation-method story stops. If the compiler helps but First-Fail-4 does not, the paper pivots to reliable persistent-state compilation and drops failure-evidence superiority. Only a positive prospective generator effect plus the required falsifiers can justify changing the working title from “Diagnosing State-Generation Variance” to the method-positive “Variance-Controlled State Compilation.”

7 DISCUSSION, LIMITATIONS, AND CONCLUSION

The scientific shift is from data selection to representation generation. The initial E2-R17 hypothesis emphasized which rejected trajectories should reach persistent learning. The closed large-sample study and selector intervention do not support a universal witness rule. The more informative observation is downstream: one persistent state is useful when frozen, yet the same updater does not reliably recreate that utility from identical evidence. This changes the primary scientific object from “which failure should we show?” to “how does evidence become persistent state?”

State-generation variance is a candidate bottleneck, not a completed variance decomposition.

The exact-evidence replay varies both the fresh state and its downstream actor evaluation, and it contains only two new updater realizations in one outcome-selected development stream. The frozen-state remeasurement partially separates actor noise, but it does not estimate population-level updater and actor variance components. We therefore use “candidate bottleneck” and reserve stronger variance claims for a prospective repeated-synthesis design.

The typed compiler is intentionally not claimed as component-level novelty. Trace-conditioned diagnosis, success/failure contrast, execution-grounded repair, and persistent knowledge accumulation already appear in SkillRevise, SkillCAT, RethinkSkill, and WikiSkill (Liu et al., 2026a; Chen et al., 2026; Liu et al., 2026b; Tang et al., 2026). The present method matters only if canonical state generation changes a controlled evidence-matched outcome. If it does not, a new vocabulary name or implementation detail does not rescue the contribution.

Generic workflow hygiene is a serious alternative. Completion, verification, and error recovery are sensible spreadsheet-agent instructions independent of any rejected trajectory. The score-only generic control is therefore load-bearing, not a decorative baseline. If a maximal generic repair state matches FF4.COMP, the evidence-conditioned diagnosis claim fails even if both beat the free-form updater.

The controlled suite is not external validity. The six failure families are generated to permit matched causal interventions and deterministic verification. They should not be interpreted as a representative sample of user spreadsheet workflows. SpreadsheetBench and later workflow-level benchmarks can test transport only after the internal mechanism survives; opening them after a negative mechanism result would be benchmark shopping.

Selected-case development remains a source of researcher degrees of freedom. The S1 stream and the semantics used to design G1–G3 were selected after historical outcomes. We treat them as a mechanistic sandbox rather than independent confirmation. The fresh bridge suite exists specifically to break this loop: its tasks, stream assignment, held-out panels, compiler vocabulary, and decision rules are frozen before bridge outcomes.

Practical implication. Persistent text state is often attractive because it is inspectable. Inspectability does not imply reproducibility. If a free-form updater can translate the same evidence into materially different persistent artifacts, then evaluation should report state identity and regeneration stability rather than only the trajectory source and final score. Content-addressed state artifacts and explicit generator comparisons are inexpensive additions to self-evolution studies even when a typed compiler is not used.

Conclusion. Our current evidence rejects a simple universal story that failed trajectories are better training signals and identifies a more specific unresolved mechanism: useful trajectory evidence can be lost or transformed unpredictably when it is synthesized into persistent skill state. A strong state remains useful when frozen, whereas byte-identical evidence does not reliably recreate its advantage through the native updater. We therefore separate evidence selection from state generation and formulate a prospective typed-compilation intervention that can falsify the state-generation bottleneck hypothesis without changing evidence or hiding stronger generic controls. The central lesson at this checkpoint is methodological and causal: in self-evolving agents, the persistent state writer should be treated as an experimental variable, not an invisible implementation detail.

REPRODUCIBILITY STATEMENT

All load-bearing controlled experiments bind task IDs, initial and learned skill SHA-256 values, trajectory evidence hashes, model identity, provider budgets, retry rules, and completion leases. Completed units are not replayed to repair scientific outcomes. The controlled bridge suite is generated deterministically from content-addressed salts, with disjoint SCREEN and VALIDATION panels. The manuscript distinguishes completed scientific results from zero-provider proposal artifacts and does not infer outcomes from training or execution progress.

AI USE STATEMENT

Generative AI tools were used to assist with hypothesis refinement, research software, literature search, adversarial methodology review, manuscript drafting, and artifact inspection. Quantitative claims are checked against persisted experiment receipts and content-addressed artifacts. The authors are responsible for the final scientific claims, citations, and submission.

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A CLAIM BOUNDARY

B FROZEN EVIDENCE SUMMARY

DeepSeek V2 Repair2. The frozen design contains 48 pairs, 96 learned states, and 1,728 held-out units. Mean MRW-WIN-C is 0.023148, median 0.013889, standard deviation 0.077970, one-sided sign-flip $p = 0.171875$, and bootstrap interval $[-0.018519, 0.065972]$. The adjudicated verdict is HOLD_MRW_UNDERPOWERED_OR_HETEROGENEOUS.

Table 2: Current manuscript claim ledger.

Status	Statement
Supported	The 48-pair rejected-witness effect is small and heterogeneous under the frozen DeepSeek V2 design.
Supported	On the selected development stream, the historical strong First-Fail persistent state remains directionally useful across two frozen-state remeasurements.
Supported	Two fresh free-form updater realizations from reconstructed byte-identical First-Fail evidence do not reproduce the historical First-Fail advantage.
Supported, local	These observations localize persistent-state generation as a candidate bottleneck that cannot be reduced to evidence availability alone in the selected development case.
Not supported	Rejected failures are universally superior learning evidence.
Not supported	Progress proximity or diagnosticity defines a validated witness-selection law.
Not supported	Updater variance has been quantitatively separated from actor variance in the population.
Not supported yet	Typed compilation improves downstream utility or reproducibility.
Not supported yet	The mechanism generalizes to untouched E3, a second backbone, or a public benchmark.

Single-case witness selector. On `e1-tsr-00`, WIN-C/First-Fail/Progress-Fail/Progress-Contrast are 13/18, 17/18, 14/18, and 14/18. The frozen verdict is `S1_SIGNAL_FAIL_STOP_NO_S2`.

Frozen-state stability. Two complete remeasurements give First-Fail versus WIN-C of 15/18 versus 14/18 and 16/18 versus 12/18. Verdict: `FIRST_FAIL_FROZEN_STATE_STABILITY_PASS`.

Exact-evidence updater replay. All reconstructed learner-visible packet hashes match the historical First-Fail evidence. Fresh updater realizations give 15/18 versus 15/18 and 11/18 versus 12/18 against fresh WIN-C comparators. The frozen adjudication is that exact-evidence updater replication failed to reproduce the strong state, leaving state-generation realization as the candidate bottleneck.

C PROSPECTIVE STAGE STATUS

M2, the deterministic G0–G3 state-semantic intervention, is frozen under an exactly-once Recovery V3 object. At this manuscript checkpoint the scientific result is not opened as a complete 72-unit analysis, and no partial effect is used in the paper. M3 is conditional on M2 passing its original gate and provider-era veto. M4, the fresh Evidence×Generator bridge, is zero-provider proposal only. M5, untouched E3 confirmation, has no authority.

The paper therefore uses future-stage descriptions only as falsification contracts. They are not evidence and are not included in the abstract as positive method results.

D BRIDGE SUITE CONSTRUCTION

The fresh bridge suite uses new controlled blocks 7–9 and has zero task-ID overlap with the earlier 378-task controlled suite. Blocks 7 and 8 provide update candidates; block 9 provides held-out candidates. From 162 deterministic candidates, hash-based selection freezes 96 update tasks in 12 eight-task streams, 12 SCREEN held-out tasks, 12 disjoint VALIDATION held-out tasks, 12 update integrity reserves, and 30 held-out integrity reserves. Each SCREEN/VALIDATION stream set contains one stream per failure family; each held-out panel contains two tasks per family. Reserved tasks may replace only a pre-execution file-integrity failure, never a model failure or unfavorable scientific outcome.

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E WHY THE PAPER DOES NOT REUSE THE OLD TEMPORAL-SKILL MANUSCRIPT

The earlier E2 temporal-skill audit studied a different scientific object: comparator and integration-surface attribution for time-conditioned retrieval procedures. The present E2-R17 paper concerns persistent skill self-evolution from trajectory feedback. Reusing the old manuscript’s abstract, results, or framing would conflate two independent research lines. This draft is therefore rebuilt from the current E2-R17 evidence ledger rather than incrementally editing the historical temporal-skill text.